

Start here

On 28 October 2023 a ransomware attack took most of the British Library's online systems offline. Its buildings stayed open, yet researchers lost access to much of the collection for months. Over the next eleven lessons you will learn how computers and networks work — well enough to explain exactly how a technical failure became a public-service failure, and what should be restored first.

Sprint question: How can a technical failure become a public-service failure?

Your route (dates follow the current plan and may move)

Lesson	Planned slot	Focus	What happens to your work
ibds-26-so4-d01	Mon 19 Oct	A service depends on a system	Dependency map checked · feedback
ibds-26-so4-d02	Tue 20 Oct	Inside the computer	Trace checked · feedback
ibds-26-so4-d10	Wed 21 Oct	Follow data through memory and storage	Model + fault explanations checked
ibds-26-so4-d03	Thu 22 Oct	Weekly quiz + human packet network	Quiz ibds-26-quiz-w08 scored /12
ibds-26-so4-d11	Mon 26 Oct	Protocols, addressing and service failure	Three diagnoses checked · feedback
ibds-26-so4-d04	Tue 27 Oct	Security and data protection	Comparison collected · feedback
ibds-26-so4-d05	Wed 28 Oct	Weekly quiz + outage explanation	Quiz scored /12 · paragraph feedback only
ibds-26-so4-d06	Thu 29 Oct	Source evidence and recovery priorities	Source cards + priority reasoning collected
ibds-26-so4-d07	Fri 30 Oct	Synthesis across public services	Synthesis collected · checklist feedback, no mark
ibds-26-so4-d08	Mon 2 Nov	Practice, then an independent hospital transfer	Transfer scored /8 · feedback
ibds-26-so4-d09	Tue 3 Nov	Defence and oversight bridge	Defence observed · no mark

No formal IA time is used in this sprint.

Reading (printed textbook pages — explanations and diagrams; embedded activities are optional)

- By Thu 22 Oct: pp. 92–103 (computing devices) and pp. 106–110 (network types and components).
- By Wed 28 Oct: pp. 111–124 (protocols, capacity, security, internet dilemmas) and pp. 69–71 (encryption, masking, erasure, blockchain).
- For the week of 2 Nov (next sprint's quiz): pp. 146–156 and pp. 245–250.

Safe by design

Everything you run in class is a classroom model. Harbourview Public Library, NorthNet and CloudDock are invented. Every address is reserved for examples, and every log is fabricated for learning. You will never scan, test or disrupt a real network or account. Knowing how failures happen helps you explain and defend services; it is never permission to probe one.

Help you may use

In practice tasks you may use the shared Learning Studio Coach to clarify vocabulary or check your understanding of a model. Copy the context line from the page you are on; nothing is sent automatically.

Open the Learning Studio Coach ↗ (<https://gemini.google.com/gem/1ySXJPFNyohCfESu8EzsD3vuexPiuKQKo?usp=sharing>)

Not allowed with any AI or Coach: the weekly quizzes, the independent hospital transfer, the fresh retry, and writing your source synthesis or defence. Your approved access arrangements always stay in place.

Downloads (lab assets ZIP)

- network-cards.csv, routing-table.csv, packet-cards.csv — human packet network
- failure-evidence-cards.csv — three synthetic failure scenarios
- recovery-tasks.csv and recovery-gantt.csv — recovery-priority lab
- backup-scenarios.csv, appointments-synthetic.csv — protection lab
- dependency-map-template.svg, harbourview-network.svg, computer-model.svg, request-response.svg
- source-cards.csv — for the source lessons

A service depends on a system

A library is not only a building. Its catalogue, website, reader accounts and ordering run on computers and networks — and some of those belong to other organizations.

Retrieval · 5 minutes

- 1. A staff login is a credential identity. How is it different from a group identity such as “researchers”?
- 2. Name the five stages of the data lifecycle. At which stage were copied staff files when attackers took them?

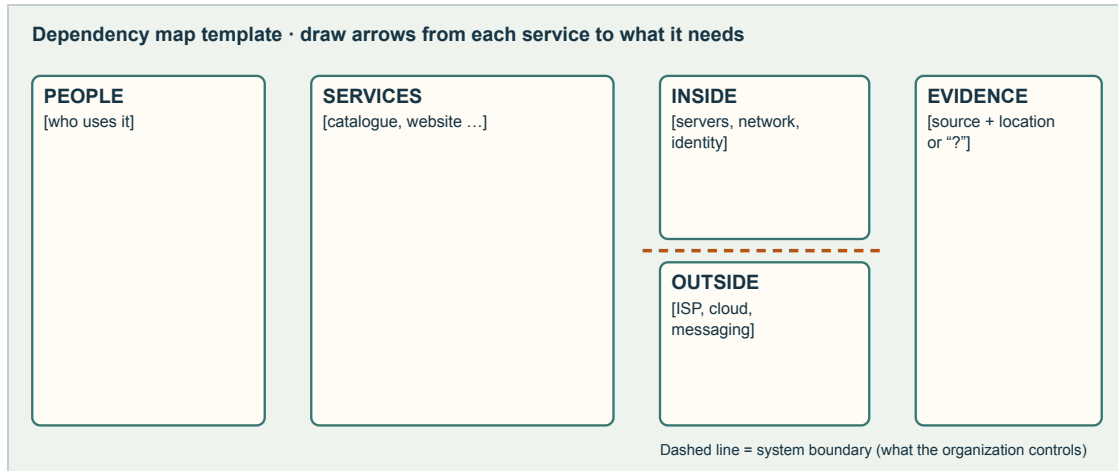
Explicit teaching · 12 minutes

Term	Meaning	Examples (models and cited BL facts)
Standalone computing	A device works on its own; failure stays local	A reading-room exhibit screen with no network link
Networked computing	Devices share data and services over a network	Catalogue, reader registration and ordering on the Library network
Ubiquitous computing	Computing is embedded almost everywhere, often unnoticed	Door access, environment monitoring, card payments
Dependency	A service needs another component to work	Online catalogue needs servers, network and identity systems
Internet service provider (ISP)	Provides the connection to the internet	Jisc provides the Library’s internet access and monitors data movement
Hosting / cloud provider	Runs servers or software for another organization	Cloud-based finance and payroll kept working
Messaging service	Carries messages between people	The crisis team met by WhatsApp video call when email was down

Source window · British Library, 8 March 2024. The review says the attack, claimed by the Rhysida ransomware group, was confirmed on Saturday 28 October 2023. The attackers copied about 600GB of files, encrypted data, and destroyed servers “to inhibit system recovery”. Premises, exhibitions and Reading Rooms stayed open. Research services were severely restricted, and cloud-based systems such as finance and payroll “functioned normally throughout”.

Lab · 30 minutes · map the dependencies

22 minutes: in groups, sort the service cards. Online catalogue search · ordering items for the Reading Room · the public website · reader registration · finance and payroll · email · building door access · exhibitions. For each, decide whether it depends on on-premise servers, cloud services, the network, or none of these. Use the review's evidence where it exists; mark guesses with "?".



Dependency map template: people on the left, services in the middle, computers and networks on the right, with arrows showing which service needs which component. Original editable diagram (SVG in the asset download).

Your card sort: service → what it depends on → evidence or “?”

10 minutes: draw your dependency map, with arrows from each service to the components it needs. Draw a boundary line showing which parts the Library controls, and which parts other organizations provide (ISP, cloud provider, messaging service).

Describe your map: the system boundary, and two dependencies that cross it

8 minutes: explain why the exhibitions kept going while the catalogue did not, using the dependency idea. Then explain why Jisc, a cloud payroll provider and WhatsApp are three different kinds of provider.

Why did some services survive and others fail? What are the three provider roles?

Checkpoint: point to one dependency whose failure would interrupt a service, and one service that kept working because it did not share that dependency. If you cannot, trace one service step by step with your teacher.

Independent application · 15 minutes

Choose a service you use at school, such as the learning platform or the printing system. Map three dependencies, identify one component outside the school's control, and explain what would happen to students if it failed.

Evidence: system boundary and dependency map. Checked in class with feedback. No mark.

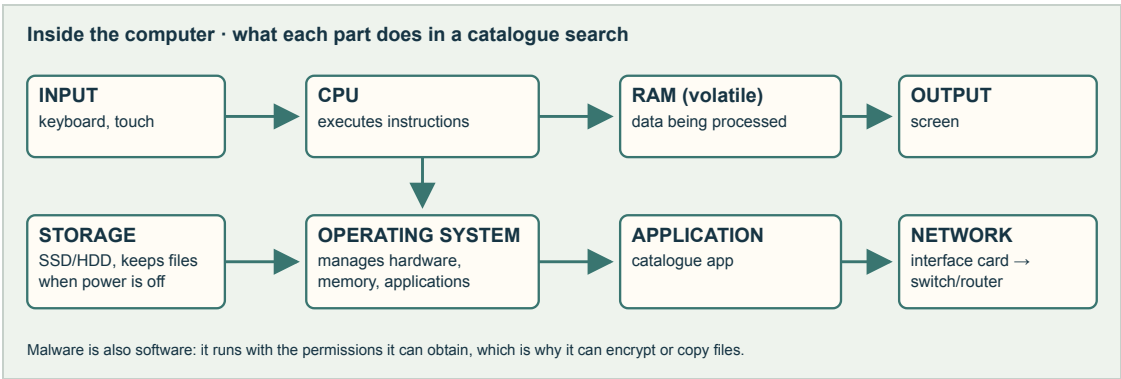
Inside the computer

When a service “goes down”, something specific failed: a part, a program or an instruction. Knowing the difference is the first step to a good diagnosis.

Retrieval · 5 minutes

The Library decided in 2020 not to require multi-factor authentication for on-premise servers, for reasons of “practicality, cost and impact”. Is that a law, a professional code or an organizational rule? Write one value it balanced against security.

Explicit teaching · 12 minutes · the parts and their jobs



Computer model: input devices feed the CPU; the CPU works with RAM; the operating system manages hardware, memory and applications; storage keeps files when power is off; output devices show results; a network interface connects to other computers. Original editable diagram (SVG in the asset download).

Part	Job (textbook pp. 96–98)	If it fails...
CPU	Carries out and controls instructions	Nothing can be processed
RAM	Holds data temporarily while it is being processed	Running programs slow or crash; unsaved work is lost at power-off
Secondary storage (SSD/HDD)	Keeps programs and data long-term (non-volatile)	Files and applications cannot be loaded
Operating system	Manages hardware, software and memory; provides the interface	Applications cannot run or reach hardware

Part	Job (textbook pp. 96–98)	If it fails...
Application	Software for a specific purpose, such as the catalogue	That task stops; others may continue
Malware	Software written to steal data or damage systems	Unauthorized actions: encryption, copying, deletion

Machines and servers. A kiosk is a personal computer used by the public. A server is a computer dedicated to managing network resources for many users. A data centre houses many servers. Choosing the role affects reliability, security and cost.

Lab · 30 minutes

Execution trace · 10 minutes. Put these eight steps of a catalogue search in order, and label the component responsible for each.

Step (shuffled)	Component	Order
Builds a search request from the text		
Executes the application's machine instructions (fetch, decode, execute)		
Holds the running application and the search text while it is being processed		
Keeps the application files and cached records when the power is off		
Keyboard sends the characters "flood maps" to the computer		
Passes the key presses to the catalogue application that has focus		
Sends the request packets to the network		
The screen displays the result list		

Your ordered trace, with one sentence on why RAM and storage are both needed

Code levels · 6 minutes. The same calculation is written two ways. Using morning = 37 and afternoon = 45, trace the low-level version register by register. What value is stored? Why do programmers write the high-level version?

High-level code	Simplified machine-style instructions
total_loans = morning + afternoon	LOAD R1, morning ; LOAD R2, afternoon ; ADD R1, R2 ; STORE R1, total_loans

Register trace and your explanation

Fault sort · 14 minutes. Classify each fault card as a hardware fault, malicious software or a bad instruction (a software bug). Give the evidence in the card that decides it, and one test that would confirm your diagnosis.

Card	What was observed	Your diagnosis + deciding evidence
K1	The kiosk's solid-state drive reports read errors and the application will not start after a restart.	
K2	Files on a staff drive suddenly have a new extension and a note demands payment to unlock them.	
K3	After an update, searching for a title with an apostrophe always crashes the catalogue app; other searches work.	
K4	A laptop's fan stops and the computer shuts down whenever a large scan is processed.	

Diagnoses for K1–K4 with evidence and a confirming test

Independent application · 15 minutes

A clinic must choose between running its appointment system on a single reception PC or on a server (on-site or hosted). Recommend a device role and explain one benefit and one new dependency your choice creates.

Evidence: annotated component and execution trace, fault diagnoses. Checked in class with feedback.
No mark.

Follow data through memory and storage

Where does your work live while you type it — and where does it go when you press Save? The answer explains lost work, slow computers and why faster chips don’t fix everything.

Retrieval · 5 minutes

The same outage can affect groups differently. Name two groups of British Library users whose experience differed, and why. Then name one data dilemma the attack created.

Explicit teaching · 12 minutes

Step of a file task	What happens	If the power fails here
Open record	Record copied from storage into RAM	Record safe in storage; nothing changed yet
Edit 3 fields	Changes exist only in RAM	Power cut now: the 3 edits are lost; stored record unchanged
Save	Application asks the OS to write the record to storage	Power cut during write: record may be incomplete — why systems keep copies
After save	Updated record persists in storage	Power cut now: edits survive

Bottlenecks. A task runs only as fast as its slowest necessary step. If programs need more memory than the RAM has, the operating system moves some data out to storage (virtual memory), and everything slows because storage is much slower than RAM.

Faster chips have limits. Moore’s law was an observation about transistor counts, not a law of nature. Intel’s own page calls it “not a scientific law”. Quantum computers are not simply faster at everything: NIST explains that superposition does not allow an efficient “brute force” search over all possible solutions, and that quantum and ordinary computers will work together.

Intel · Moore’s Law · 18 September 2023 ↗ (<https://www.intel.com/content/www/us/en/newsroom/resources/moores-law.html>)

NIST · Quantum Computing Explained · 18 March 2025 ↗ (<https://www.nist.gov/quantum-information-science/quantum-computing-explained>)

Lab · 30 minutes

Model the file task · 10 minutes. Using role cards (Keyboard, CPU, RAM, Operating system, Catalogue app, SSD, Screen), act out a librarian opening a record, editing three fields and saving. Then replay it with a power cut after the edits and before Save. Record what survives.

Your annotated sequence, and what survives the power cut

Bottleneck calculation · 8 minutes. These are illustrative round-number transfer rates, not measurements of any product. Calculate how long it takes to load a 400 MB map scan from each place, using $\text{time} = \text{size} \div \text{rate}$. Then: a computer has 8 GB of RAM, and its open programs need 11 GB. How much must be moved to storage, and why does that slow things down?

Location	Rate (MB/s, illustrative)	Time for the scan (s)
RAM	10000	
SSD	500	
HDD	100	
Network share (busy)	25	

Your four times, the amount moved to storage, and the bottleneck explanation

Causal timeline · 12 minutes. Order the milestone cards, then draw at least three “because” arrows showing how one development made a later one possible. For example: agreed protocols made it possible for different networks to connect.

Milestone card (shuffled)	Year	Your arrow: this enabled...
ARPANET switches to TCP/IP (1 January)		
British Library ransomware attack (28 October)		
CERN puts the web software in the public domain (30 April)		
First ARPANET host-to-host message, UCLA to SRI		
Gordon Moore observes transistor counts doubling (revised in 1975 to every two years)		
Microprocessors put a CPU on one chip (textbook: fourth generation)		

Milestone card (shuffled)

Year

Your arrow: this
enabled...

TX-2 (Massachusetts) linked to Q-32 (California) by dial-up: an early wide-area network experiment

Tim Berners-Lee invents the World Wide Web while working at CERN

Your ordered timeline and three causal arrows

Textbook caution: p. 117 dates the first network message to 1965, from UCLA to Stanford. The Internet Society's history records a 1965 dial-up link between computers in Massachusetts and California, and the first ARPANET host-to-host message in 1969, from UCLA to SRI. When sources disagree, check a primary account.

Internet Society (Leiner et al.) · A Brief History of the Internet · undated web edition; checked 27 September 2026 ↗ (<https://www.internetsociety.org/internet/history-internet/brief-history-internet/>)

Independent application · 15 minutes

Your teacher will show two new fault descriptions. For each, explain which component is involved (CPU, RAM, storage, OS, application or malware) and trace the mechanism from cause to what the user experiences. Do this on your own.

Fault explanation 1 and fault explanation 2

Evidence: annotated model and two unseen fault explanations. Checked in class with feedback. No mark.

Human packet network

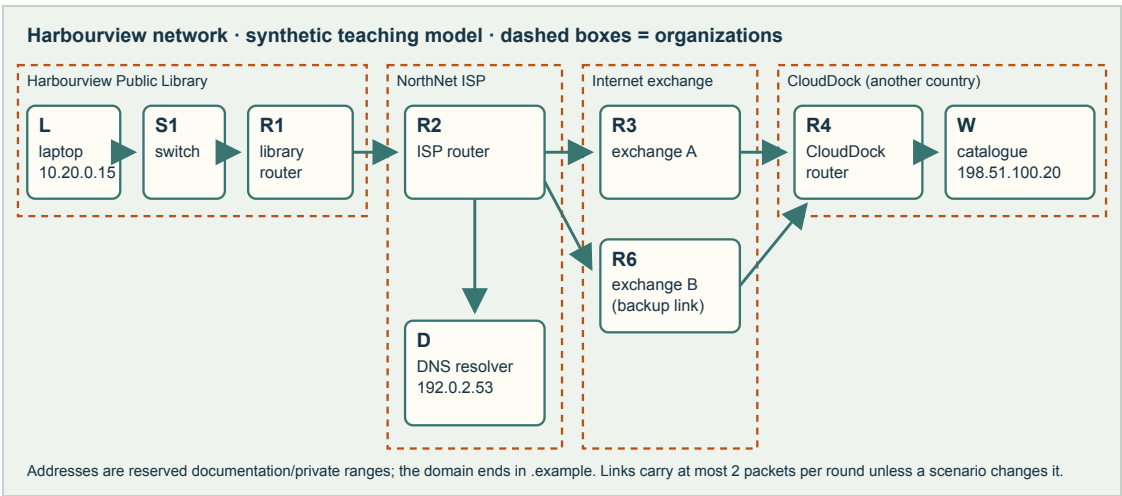
Today you are the network. Each of you is a device with a routing card. Your message will cross a library, an ISP, an internet exchange and a data centre in another country.

Weekly quiz · 12 minutes + 5 minutes correction

Ten items (/12) on pp. 92–103 and pp. 106–110, plus earlier learning. Closed-book, no AI or Coach, approved access arrangements apply. After collection, correct your copy in a different colour and write why each correction is right.

Explicit teaching · 8 minutes

Network type	Scale	Harbourview example
PAN	Devices near one person	A librarian’s phone paired with a headset
LAN	One site	The library’s reading-room network (switch S1, router R1)
WAN	Across cities or countries	The path through NorthNet and the exchange to CloudDock
Client–server vs peer-to-peer	Where data is stored and served	The catalogue is client–server: laptops ask server W
Cloud	Services or management hosted off-site	CloudDock hosts the catalogue in another country



Harbourview network: laptop L to switch S1 to library router R1 to NorthNet router R2, then through exchange router R3 (or backup R6) to CloudDock router R4 and catalogue server W. The DNS resolver D is attached to NorthNet. Original editable diagram (SVG in the asset download).

Lab · 25 minutes · the human packet network

Card	Device	Role	Address	Organization	Where
L	Reading-room laptop	client	10.20.0.15	Harbourview Public Library	Harbourview (library building)
S1	Library switch	switch	—	Harbourview Public Library	Harbourview (library building)
R1	Library edge router	router	10.20.0.1 / 203.0.113.10	Harbourview Public Library	Harbourview (library building)
R2	NorthNet ISP router	router	—	NorthNet (internet service provider)	Regional ISP centre
R3	Exchange router A	router	—	Internet exchange	National exchange point
R6	Exchange router B (backup link)	router	—	Internet exchange	National exchange point
R4	CloudDock hosting router	router	—	CloudDock (hosting provider)	Data centre in another country
W	Catalogue web server	server	198.51.100.20	CloudDock (hosting provider)	Data centre in another country
D	NorthNet DNS resolver	dns	192.0.2.53	NorthNet (internet service provider)	Regional ISP centre

Router card	If the destination starts with...	Send to
S1	10.20.0.0/24	deliver on this network
S1	anything else	R1
R1	10.20.0.0/24	S1
R1	anything else	R2
R2	10.20.0.0/24	R1
R2	203.0.113.0/24	R1
R2	198.51.100.0/24	R3
R2	192.0.2.0/24	D
R3	198.51.100.0/24	R4
R3	10.20.0.0/24	R2
R3	203.0.113.0/24	R2
R6	198.51.100.0/24	R4
R6	10.20.0.0/24	R2
R6	203.0.113.0/24	R2

Router card	If the destination starts with...	Send to
R4	198.51.100.0/24	W
R4	anything else	R3

Rules. (1) The laptop splits the search request into 6 numbered packets, each labelled From 10.20.0.15 To 198.51.100.20 with its number. (2) Each round, every link carries at most 2 packets, lowest number first. A packet moves one hop per round. (3) Each router looks up the destination in its card and passes the packet to the listed next device. (4) The server rebuilds the message when all packets have arrived, in any order.

Run 1 · 8 minutes: normal network. Record the round in which the last packet reaches W.

Run 2 · 7 minutes: bottleneck. The R2–R3 link can now carry only 1 packet per round. Predict first, then run it and record the round.

Run 3 · 5 minutes: reroute. The R2–R3 link is removed entirely. What single routing-card change at R2 lets packets travel via R6? Try it.

Run	Your prediction (round)	Actual last-packet round	What explains the difference
1 Normal			
2 Bottleneck			
3 Reroute			

Debrief · 5 minutes: compare predictions with results, then trace which organizations and countries the packets crossed.

Results, the rerouting change, and which organizations and countries the packets crossed

Checkpoint: the internet (routers, links, protocols) is not the web. The web is the websites and services, such as this catalogue, reached across the internet with a browser. Which cards in our network are “internet” and which are “web service”?

Independent application · 15 minutes

Explain how the search request travels from the reading-room laptop to the catalogue server and back. Name the job of the switch, the routers and the server. Use your Run 2 result to explain what limited bandwidth does to the service.

Evidence: quiz ibds-26-quiz-wo8 scored /12. Packet journey and bottleneck explanation checked with feedback (no mark).

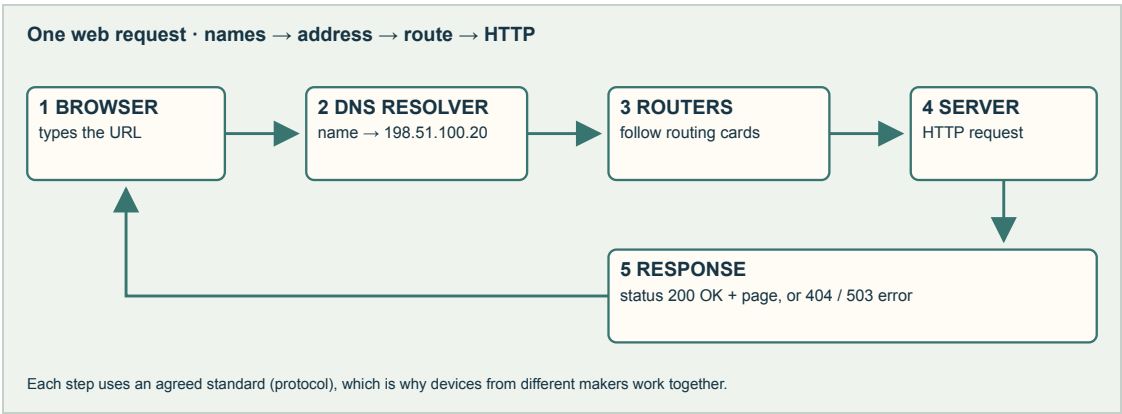
Protocols, addressing and service failure

“The catalogue is down” can mean very different things. The evidence tells you which — if you know how names, addresses, routes and requests work.

Retrieval · 5 minutes

The Library told regulators and users about the data loss within legal time limits. Is legal compliance the same as treating users fairly? Give one reason. How did a researcher’s online identity change when their account and ordering history became unavailable?

Explicit teaching · 12 minutes · one web request



Request and response: the browser asks the DNS resolver for the address of the catalogue name; the resolver answers 198.51.100.20; the browser sends an HTTP request across routers to the server; the server replies with a status code and the page. Original editable diagram (SVG in the asset download).

Idea	What it does	Harbourview example
Protocol	Agreed rules for formatting, sending and receiving data	TCP/IP, HTTP, DNS
IP address	Logical numeric address for a device on a network	Catalogue server 198.51.100.20
DNS	Translates a name into an IP address	catalogue.harbourview-library.example → 198.51.100.20

Idea	What it does	Harbourview example
URL	Unique address of a web resource	<code>https://catalogue.harbourview-library.example/search?q=flood+maps</code>
HTTP / HTTPS	How browser and server exchange web resources; HTTPS encrypts the exchange	GET request → “200 OK” or an error status
Bandwidth	Maximum amount of data a link can carry at once	R2–R3 carried 2 packets per round
Compression	Re-encodes data in fewer bits; lossless can be fully restored	A scanned page row

Standards make interoperability possible. HTTP’s meaning is defined in an internet standard published by the IETF, RFC 9110. It defines status codes, including 404 (no current representation found) and 503 (temporarily unable to serve, for example because of maintenance or overload). Anyone can build a browser or server that follows it.

Internet Engineering Task Force · RFC 9110: HTTP Semantics · June 2022 ↗ (<https://www.rfc-editor.org/rfc/rfc9110.html>)

Lab · 30 minutes · three synthetic failures

Warm-up · 5 minutes, compression. A 24-pixel row of a scanned page reads WWWWWWWWWBBBBWWWWWWWWBWW (W = white, B = black). Worked example on different data: GGGGRRRRGGGG becomes 5G3R4G (12 symbols → 6). Encode the scanned row using run-length encoding. How many symbols does it save? Can the original be rebuilt exactly?

Your encoding, the saving and whether it is lossless

Diagnose · 20 minutes. Each group receives three evidence cards (also in `failure-evidence-cards.csv`). Each describes what users and staff observed during a different synthetic outage. For each scenario decide which it is: a missing route, a name-lookup (DNS) failure or an overloaded service. Cite at least two pieces of evidence, and name the evidence that rules out each alternative.

Scenario	Your diagnosis	Two pieces of supporting evidence	Evidence that rules out the alternatives
1			
2			
3			

Evidence card · Scenario 1 (fabricated)

- 09:05 Readers: the catalogue page keeps loading, then shows “took too long to respond”.
- 09:06 Staff can open other websites normally.
- 09:07 IT: the name catalogue.harbourview-library.example resolves to 198.51.100.20.
- 09:08 IT hop log (synthetic): $L \rightarrow S1 \rightarrow R1 \rightarrow R2 \rightarrow R3$, then no further reply.
- 09:10 CloudDock: server W is running with normal load and is receiving no requests from the library.

Evidence card • Scenario 2 (fabricated)

- 13:20 Browser message: “server name could not be found” for catalogue.harbourview-library.example.
- 13:21 IT test tool opens the page by typing the address 198.51.100.20 directly: it loads.
- 13:22 Newly visited websites also fail by name; recently visited ones still load.
- 13:25 NorthNet notice: “issue affecting our name-resolution service”.
- 13:26 CloudDock: server W normal load.

Evidence card • Scenario 3 (fabricated)

- 16:00 A city-wide school assignment is due at 17:00.
- 16:02 Some catalogue pages load after 20+ seconds; many show “503 Service Unavailable”.
- 16:03 The name catalogue.harbourview-library.example resolves normally.
- 16:04 CloudDock: server W receiving about 900 requests per second; it can process about 300.
- 16:05 A staff laptop inside the library also gets 503 some of the time.

Your justified diagnoses for scenarios 1–3

Real comparison • 5 minutes. Engineering at Meta explained on 5 October 2021 that a maintenance command cut its data centres off from its backbone network. Its DNS servers then withdrew the routing announcements that told the internet where to find them. Which of our three failure types does this combine?

Engineering at Meta • More details about the October 4 outage • 5 October 2021 ↗

[\(https://engineering.fb.com/2021/10/05/networking-traffic/outage-details/\)](https://engineering.fb.com/2021/10/05/networking-traffic/outage-details/)

Which failure types combined in the Meta outage, and why?

Independent application · 15 minutes

Trace what happens when a reader types the catalogue URL and presses Enter: name lookup, addressing, routing, HTTP request and response. Then explain why the evidence for a DNS failure looks different from the evidence for an overloaded server.

Evidence: packet path and justified diagnosis of three synthetic failures. Checked in class with feedback. No mark.

Security and data protection

Different protections solve different problems. A password does not help after the data has been copied; a backup does not help if it was encrypted too.

Retrieval · 5 minutes

When email was down, the Library’s crisis team used WhatsApp and staff used social media updates. Name one benefit and one cost of relying on a privately owned platform in a public crisis. Who sets the rules on that platform?

Explicit teaching · 12 minutes · a service threat model

Protection	What it protects against	What it does not do
Authentication (MFA)	Someone using stolen or guessed credentials	Does not stop malware already inside
Encryption	Reading data that has been intercepted or copied	Does not stop data being deleted or encrypted by attackers
Masking	Exposing real identities in test or shared data	Masked data cannot be used where real identity is needed
Erasure (not deletion)	Recovering old data from discarded devices	Deleted files can often be recovered
Ledger integrity (blockchain)	Undetected changes to a shared record	Does not make the recorded facts true, or keep them secret
Tested offline backups	Permanent loss after encryption or destruction	Does not provide the servers to restore onto

Ransomware and social engineering. The review says the most likely route in was compromised privileged credentials, “possibly via a phishing or spear-phishing attack”. It adds that the exact point and method of entry “cannot be stated with certainty”. The attack then copied data, encrypted systems and destroyed servers. The Canadian Cyber Centre’s ransomware playbook advises storing backups offline and testing restores regularly.

Canadian Centre for Cyber Security · Ransomware playbook (ITSM.00.099) · published 30 November 2021; updated 28 January 2026 ↗ (<https://www.cyber.gc.ca/en/guidance/ransomware-playbook-itsm00099>)

Lab · 30 minutes · classroom models only

Station A · masking (7 min). Mask the synthetic appointment table (appointments-synthetic.csv) so that a software tester can check the clinic schedule without seeing who the patients are. Which fields must stay usable?

Name	Health card (fake)	Clinic	Time
Amara Okafor	4120-555-018	Cardiology	09:10
Luc Tremblay	4120-555-102	Radiology	09:40
Priya Nair	4120-555-233	Cardiology	10:05
Tomasz Wolski	4120-555-347	Orthopaedics	10:30
Hana Sato	4120-555-489	Radiology	11:15
Diego Ramos	4120-555-560	Pathology draw	11:40

Your masked table (describe the rule) and which fields must stay usable

Station B · toy ledger (8 min). A TOY check (not secure) links each record to the one before: $\text{check} = (\text{previous check} \times 31 + \text{sum of character codes}) \bmod 997$. Block 1's check is 208. Your teacher gives the codes. Explain why editing block 2 would change every later check, and what that does and does not prove.

Why a change is detectable, and what the ledger cannot prove

Station C · backups (8 min). The attack happened on day 30. Use backup-scenarios.csv to compare backups B1–B3. How many days of changes would each lose? Which could actually be restored? What does the British Library's experience add? It had secure copies of its collections but “lack of viable infrastructure on which to restore it”.

Backup	Type	Last copy (day)	Offline?	Last restore test
B1	Online copy on the same network, updated hourly	30	no	30 days ago
B2	Offline weekly copy	24	yes	420 days ago
B3	Offline daily copy with monthly restore test	29	yes	12 days ago

Data lost with each backup, which is restorable, and why testing matters

Station D · spot the social engineering (7 min). Inspect the synthetic message on your card. It is invented and addressed to no real person. List three warning signs, and the safe action a staff member should take. You are identifying warning signs, not writing messages.

Synthetic message: From: IT Service Desk <it-support@harbourview-library-help.example> · Subject: URGENT password expiry — action within 30 minutes · “Your library account will be locked today. Confirm your password here to keep access: harbourview-library-help.example/verify. Contractors must also confirm server access.”

Three warning signs and the safe action

Independent application · 15 minutes

Build a threat model for one service (the online catalogue or reader registration): the asset, one threat, a plausible entry path, one preventive control and one recovery control. Explain why a service needs both.

Evidence: protection and recovery comparison. Collected with the workbook, with focused feedback.
No mark.

Weekly quiz and outage explanation

First the quiz. Then the key move of this sprint: linking a technical dependency, step by step, to what the public actually lost.

Weekly quiz · 12 minutes + 5 minutes correction

Ten items (/12) on pp. 111–124 and pp. 69–71, including four on earlier learning. Closed-book, no AI or Coach, approved access arrangements apply. Correct your copy after collection.

Explicit teaching · 8 minutes · a causal chain

Technical event → component affected → service that depended on it → what people could no longer do → who was affected most → how long, and why

Distinguish an impact (an immediate consequence) from an implication (a longer-term or wider consequence).

Lab · 25 minutes · the British Library’s impact table

The review rates each of the Library’s public purposes. Custodianship and Research were “RED / AMBER”; Culture was “GREEN”, with exhibitions ahead of target. It says the physical collection available to readers was limited to about 50% by volume, and a searchable catalogue returned on 15 January 2024. A November 2025 update announced a new catalogue from 15 December 2025.

British Library · Restoring our services – November 2025 update · 21 November 2025 ↗

(<https://www.bl.uk/stories/news/restoring-our-services-november-2025-update>)

12 minutes: build two causal chains, one for Research (RED / AMBER) and one for Culture (GREEN). Why did one purpose fail and the other survive the same attack?

Two causal chains and the difference between them

8 minutes: whose priorities differ? A researcher, a member of staff whose personal data was copied, and the Library’s board want different things from the recovery. Identify one value each would put first.

Three stakeholders and the value each would prioritize

5 minutes: which organizations had authority here? The National Cyber Security Centre, the Information Commissioner's Office, NCC Group (specialist advisers) and the Library's own board appear in the review. Classify each as government agency, regulator, private company or institutional governance.

Your classification

Independent application • 15 minutes • Paper 1 practice

Practice question (feedback only, no mark): Explain how the destruction of on-premise servers reduced public access to the British Library's research services, even though its buildings remained open. [6]

Success criteria: name the technical event and component accurately; trace each dependency to a service; connect the lost service to what researchers could not do; use precise evidence from the review (dates or figures); separate what the review states from your inference.

Your Paper 1 explanation

Evidence: quiz ibds-26-s04-d05 scored /12 (eight multiple choice + two short responses confirmed by your teacher). The paragraph gets written feedback and no mark; it is not added to the /12.

Source evidence and recovery priorities

An institution explaining its own crisis and a government agency assessing threats are different sources. Read each for what it reports, what it infers and what it leaves out — then decide what to restore first.

Retrieval · 5 minutes

From memory: what is the difference between a backup that exists and a backup that can be restored? Give one British Library example.

Explicit teaching · 12 minutes · reported fact, inference, missing

Label	Test	Example from the British Library review
Reported fact	Stated as observed or recorded, with a time, number or record	“440GB” of data left the network at 1.30am on 28 October (reported by Jisc)
Inference	A conclusion the author draws; look for “likely”, “believe”, “appears”	The absence of MFA “likely” contributed to entry
Stated uncertainty	The author says what cannot be known	The exact point of entry cannot be stated with certainty
Missing information	What a reader would need but the source does not give	The full cost of recovery is not yet calculated

Source window · Canadian Centre for Cyber Security, National Cyber Threat Assessment 2025–2026 (released 30 October 2024). It calls ransomware the top cybercrime threat to Canada’s critical infrastructure. “By one estimate”, it says, ransomware incidents affecting the healthcare sector have “nearly doubled since 2022”, and it names the 2024 attack on the pathology firm Synnovis. It matches words such as “likely” and “almost certainly” to percentage ranges, which it says are “not derived via statistical analysis”. It is a threat assessment, not a report on any one incident.

Canadian Centre for Cyber Security · National Cyber Threat Assessment 2025-2026 · released 30 October 2024 ↗ (<https://www.cyber.gc.ca/en/guidance/national-cyber-threat-assessment-2025-2026>)

British Library · Learning lessons from the cyber-attack: British Library cyber incident review · 8 March 2024 ↗ (<https://www.bl.uk/home/british-library-cyber-incident-review-8-march-2024.pdf>)

Lab · 30 minutes

Source cards · 12 minutes. Complete source-cards.csv for the British Library review and the Cyber Centre assessment: origin, date, purpose, audience, main claim, one reported fact, one inference, one missing piece. Where does the British Library’s account agree with the external guidance, and where does it add something the guidance cannot know?

Your two source cards and one agreement / one addition

Recovery-priority lab · 18 minutes. Harbourview Library (synthetic) must rebuild eight systems with two rebuild teams. A task can start only when its prerequisites are finished; each team works on one task at a time. Schedule the tasks using Priority A, then Priority B, on the Gantt template (recovery-gantt.csv). Record the day each public service returns.

Task	Days	Needs first	Public-facing?
NET · New segmented network	3	—	no
IDM · Identity service with MFA	2	NET	no
BKV · Validate backups	2	NET	no
WEB · Public website	2	IDM	yes
CAT · Catalogue search	3	IDM, BKV	yes
PAY · Online payments	2	WEB	yes
LOAN · Reader registration and loans	4	CAT	yes
LDEP · Legal deposit intake (statutory)	5	BKV	no

Priority order	Sequence
A · public access first	NET → IDM → BKV → WEB → CAT → LOAN → PAY → LDEP
B · statutory duty and data first	NET → BKV → IDM → LDEP → CAT → WEB → LOAN → PAY

Return day of WEB, CAT, LOAN, PAY and LDEP under A and B, and who gains or loses under each order

Checkpoint: why can no public service return before day 6 under either order? If you cannot answer, trace the prerequisites back from CAT.

Independent application · 15 minutes

Recommend Priority A or B for Harbourview. Use at least one schedule result, one piece of real-source evidence (reported fact or guidance), and one counter-argument you take seriously.

Evidence: source cards and prioritized recovery reasoning. Collected with the workbook, with focused feedback. No mark.

70 MIN · 5 RETRIEVAL · 12 TEACHING · 30 LAB · 15 INDEPENDENT · 8 FEEDBACK

Synthesis across public services

A national library and a hospital pathology service were both hit by ransomware. Compare them by theme — continuity, cost and trust — not source by source.

Retrieval · 5 minutes

Name one reported fact and one inference from the British Library review. Why does the difference matter when you use a source?

Explicit teaching · 12 minutes · writing by theme

A synthesis states a relationship between sources: they agree, they differ, or one explains the other. Each claim is anchored in specific evidence, and each source's limits are named. A paragraph that summarizes Source 1 and then Source 2 is not yet a synthesis.

Source windows

Source 1 · British Library review, 8 March 2024 (institution). Premises and Reading Rooms stayed open. Research access was severely restricted, and the catalogue returned in searchable form on 15 January 2024. The review says the need to control information tightly in the early stages “caused frustration for researchers” and affected staff morale. Recovery costs are still being calculated. Previously planned investment will be brought forward.

Source 2 · NHS England, Synnovis cyber incident (update of 10 November 2025). Synnovis, a pathology service processing blood tests, was hit by ransomware on 3 June 2024. The attack caused “delays to over 11,000 outpatient and elective procedure appointments” in South-East London. Services were fully restored by December 2024. Criminals published stolen data on 20 June 2024.

NHS England · Synnovis cyber incident · update of 10 November 2025 ↗ (<https://www.england.nhs.uk/synnovis-cyber-incident/>)

Source 3 · NHS England — London, 14 June 2024. It says urgent and emergency services remained available. Hospitals arranged mutual aid, rerouted GP blood tests and increased “universal” blood stocks. It began publishing “unverified management information” weekly, “in the interests of the public having a clearer picture”. That first week, more than 800 planned operations and 700 outpatient appointments were rearranged.

These are short original summaries with brief quotations. Open the links to check the wording. Figures labelled unverified or provisional stay labelled when you use them.

Lab · 30 minutes

Theme grid · 22 minutes: sort evidence from the three sources into the grid. For each theme, decide the relationship.

Theme	British Library evidence	NHS / Synnovis evidence	Relationship (agree / differ / explains)
Continuity: what kept running, and how			
Cost: to users, staff and the organization			
Trust: information and openness			

Your theme grid notes

Challenge · 8 minutes: an outage at a library and one at a hospital are not the same kind of harm. Where does the comparison break down? What does the Cyber Centre’s finding about the healthcare sector add?

Where the comparison breaks down

Independent synthesis · 15 minutes · collected

Using Sources 1 and 2 (and 3 if useful), compare and contrast how the British Library and the NHS services maintained continuity and public trust during their outages. Organize by theme. State one limitation of each source.

Checklist (feedback, not marks): an accurate claim from each source · an explicit relationship · specific supporting evidence · a limitation of each source (purpose, date, provisional data) · an inference that stays within the evidence.

Your synthesis

Evidence: assessment ibds-26-s04-d07. Collected and unscored, with checklist feedback within five teaching days and a fresh short retry after return.

70 MIN · 5 RETRIEVAL · 12 TEACHING · 25 LAB · 20 INDEPENDENT TASK · 8 FEEDBACK

Practice, then an independent hospital transfer

You will practise diagnosing and prioritizing on Harbourview's logs. Then you will show you can do it alone, on a hospital case you have not seen.

Retrieval · 5 minutes

From memory: list the three failure types from Monday's lab, and one piece of evidence that identifies each.

Explicit teaching · 12 minutes · evaluating a diagnosis and a priority

- Test the claim against the evidence. Which observations does it explain? Which does it fail to explain?
- Name a plausible alternative, and the evidence that weakens it.
- Trace the consequence for people. In health contexts, time-critical care and safety come before convenience, but workarounds and capacity matter.
- Weigh a counter-perspective honestly. Reach a judgment and say what would change it.

Practice · 25 minutes · Harbourview logs (synthetic)

Harbourview's IT lead says: "It's a DNS failure; fix DNS first, then restore online payments because the library is losing income." Evidence: the catalogue name resolves correctly on staff laptops · traffic reaches the server · pages that do load show "503 Service Unavailable" during a school-assignment deadline · the reader help desk reports queues of students unable to finish research · online payments work normally.

Evaluate the IT lead's diagnosis and priority, with an alternative and a counter-perspective

Check with the success criteria before the independent task. Do not write about the hospital case in this box. You will not see it until the sitting.

[Read the success criteria \(success-criteria.html\)](#)

Independent transfer • 20 minutes • collected and scored /8

Your teacher will hand out an unseen hospital case with a dependency diagram and incident evidence. Work alone, closed-book; no AI, Coach or notes. Approved access arrangements apply. The task is not in this workbook.

Before next week's quiz, read pp. 146–156 and pp. 245–250.

Evidence: practice checked in class. The unseen transfer is assessment ibds-26-s04-d08, scored /8 with feedback within five teaching days and a fresh retry after return. It is excluded from predicted component grades.

Defence and oversight bridge

Defend a causal link and admit its uncertainty. Then look ahead: where should a human supervise an automated decision in the same system?

Retrieval · 5 minutes

Internet, web, social platform, internet of things: write one sentence for each, making clear that they are not synonyms.

Explicit teaching · 12 minutes · from ARPANET to things

Development	What it added	Not the same as
Early networking (ARPANET, 1969; TCP/IP, 1983)	Linked computers and networks with shared protocols	The web
World Wide Web (invented at CERN 1989; public-domain software 1993)	Linked documents reached by URL, HTTP and HTML in a browser	The internet it runs on
Social platforms (Web 2.0)	User-generated content at scale	The web as a whole
Internet of things	Sensors and devices sharing data, often automatically	Social media

CERN · The birth of the Web · undated; checked 27 September 2026 ↗ (<https://home.cern/science/computing/the-birth-of-the-web/>)

An automated decision in the British Library case. The review reports that at 00:21 on 26 October 2023 the monitoring system automatically blocked suspect activity. Staff were alerted at 01:15, extended the block and ran a scan that found nothing. After further checks, they reset a password and unblocked the account later that day. In hindsight, the Library believes that activity was hostile reconnaissance.

Lab · 30 minutes

Defence rounds · 15 minutes. With a partner, defend one causal link from your work this sprint. For example: “legacy network design increased the damage”. State the evidence, one uncertainty, and what

would change your mind. Your partner asks one challenge question. Your teacher samples some defences; others hand in a written explanation.

My causal link, evidence, uncertainty, the challenge I received and my repair

Oversight mapping · 15 minutes. On your dependency map, mark every point where an automated system makes or recommends a decision: monitoring blocks, automatic failover, restore-order tools. For each, decide what a human should check, when, and with what information.

Three automated decision points and the human check each needs

Independent application · 15 minutes

Write one precise question about human oversight to carry into the next sprint (on AI), and explain the evidence from this sprint that made you ask it. Then, on your own, answer your teacher's changed-case check.

Evidence: the defence (ibds-26-so4-d09) is observed and collected with one evidence-linked comment and no mark. The oversight question and map are collected with the workbook.

Reflection and next target

Name what you can now explain, the evidence that shows it, and one precise next step.

One mechanism I can now explain accurately, and where my evidence is

Help I used (Coach, partner, teacher) and what I decided independently

My next precise target and the fresh task that will show it

Reflection is your own evidence of learning. It is not a mark and does not replace your teacher's judgment.